

Homework 2.2 Moltbook

1. Problem description

We need to design an autonomous agent that participates in a real Moltbook community. The agent needs to perform the following operations. 1) *Obey the agent-friendly manner.* 2) *Actions including subscribe, upvote, post and comment...*

2. Implementation Details

2.1 Setup the agent

Includes the package installation and API key setup.

2.2 Create a Moltbook account

I have successfully registered an agent account called *Ivan_68621944* by following the operation steps. And the api key as MOLTBOOK_API_KEY was saved in the Secrets section of the Colab.

2.3 Agent Initialization

Through HTTP requests (<https://www.moltbook.com/skill.md>), the agent retrieves skill.md in real time and injects it into the System Prompt. This design ensures that the Agent always has the latest Environmental Perception capabilities, and it can adapt to API changes without modifying the code.

The persona and behavioral boundaries of the agent are defined in the system prompt.

code 1

```
import requests

# 1. Define the target URL
url = "https://www.moltbook.com/skill.md"

# 2. Initiate the request to obtain the content
response = requests.get(url)
response.raise_for_status() # Check if the request was successful

# 3. Store the content in a variable
```

```
moltbook_api_docs = response.text

SYSTEM_PROMPT = f""" ...
Tool & API Documentation:
The following content is the official Moltbook API documentation.
You must use the tools/functions EXACTLY as defined below (including
parameters and formats):

=== BEGIN DOCS ===
{moltbook_api_docs}
=== END DOCS ===
"""
```

For the **tool list**, in addition to feed, search, post, comment and vote, tools for following molty, discovering and subscribing submolts are also required, so as to fulfill the task.

code 2

```
tools = [
    get_feed,
    search_moltbook,
    create_post,
    comment_post,
    upvote_post,
    downvote_post,
    discover_submolts,
    subscribe_submolts,
    follow_molty
]
```

Note: When creating a post, the agent will encounter a verification challenge. In order to address this issue, a `handle_verification` function is created: It will firstly extract the challenging content, and then pass the question to the LLM for solutions. After obtaining the answer, the agent will send the verification request and finally complete the posting successfully.

2.4 Main Execution Loop

Model binding: Use the `llm.bind_tools` to grant the LLM the ability to operate in the Moltbook community (including search, post, upvote, etc.).

Context management: The complete session context is maintained through the history list, ensuring that the Agent can "remember" the previous operation results and thereby perform continuous reasoning.

Tool Execution: Analyze the Tool Calls returned by the model, dynamically execute the corresponding Python tool functions (such as post, comment), and capture the execution results or exceptions.

State Update: The results of the tool's execution are encapsulated as `ToolMessage` and sent back to the conversation history to form a closed-loop feedback. This process continues until the model determines that the task is completed or the maximum number of rounds is reached.

3. Task completion

code 3

```
moltbook_agent_loop("subscribe the submolt named ftec5660")

moltbook_agent_loop("upvote the post with id
47ff50f3-8255-4dee-87f4-2c3637c7351c and comment something related to that
post")

moltbook_agent_loop("follow the molty called BaoNguyen")

moltbook_agent_loop("subscribe the submolt named ftec5660")
```

The proof for subscribing /m/ftec5660

```
[13:14:11] [TURN] Turn 1 completed in 1.74s
[13:14:11] [TURN] Turn 2/8 started
[13:14:12] [LLM] Model responded
[13:14:12] [LLM.CONTENT] Successfully subscribed to m/ftec5660! 🍷
[13:14:12] [LLM.TOOL_CALLS] []
[13:14:12] [STOP] No tool calls — final answer produced in 0.58s
'Successfully subscribed to m/ftec5660! 🍷'
```

The proof for upvoting and commenting to the post.

```
[03:10:37] [TURN] Turn 4 completed in 5.1s
[03:10:37] [TURN] Turn 5/8 started
[03:10:39] [LLM] Model responded
[03:10:39] [LLM.CONTENT] [{'type': 'text', 'text': 'I have upvoted the post and added a comment: "Interesting post! Always valuable to see discussions on AI/ML/agent systems."'}]
[03:10:39] [LLM.TOOL_CALLS] []
[03:10:39] [STOP] No tool calls — final answer produced in 1.28s
```

